Learning Evidence

Shift Cipher

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

← alphabet

Encryption ⇒ $C = E(p) = p + k \pmod{26}$

Decryption ⇒ $P = D(c) = p - k \pmod{26}$

- Example encrypting ⇒ C O M I N G ($k = 9$)

2 14 12 8 13 6

+9 ⇒ 11 23 21 17 22 15

mod 26 ⇒ 11 23 21 17 22 15

L X V R W P

- Decrypting this ⇒ L X V R W P ($k = 9$)

11 23 21 17 22 15

−9 ⇒ 2 14 12 8 13 6

mod 26 ⇒ 2 14 12 8 13 6

C O M I N G

Affine Cipher

similar to shift cipher, slightly advanced ⇒

Encryption ⇒

$c = E(p) = ap + b \pmod{26}$

$y = ax + b \pmod{26}$

X use keys where $\gcd(a, 26) > 1$

$2x + 6 \equiv 8 \pmod{26}$

when $x = 1$ & $x = 14$

⇒ only 12 'a' values to use.

Decryption ⇒

need to find the modular inverse

$p = D(c) = (a^{-1}) * (c-b) \pmod{26}$

$x = \frac{y-b}{a} \pmod{26}$

* Example: Encrypt ⇒

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

H E A V Y $E(p) = 15p - 2$

7 4 0 21 24   ⇒ $y = 15x - 2 \pmod{26}$

103 58 −2 313 360 ⇐ $y = 15x - 2$

25 6 24 1 20 ⇐ $y = 15x - 2 \pmod{26}$

Z G Y B U

∴ HEAVY = ZGYBU

* Example: Decryption ⇒ ZGYBU where $E(p) = 15p - 2$

equation ⇒ $y = 15x - 2 \mod 26$

$y + 2 = 15x \mod 26$

$\frac{y+2}{15} = x \mod 26$

$(15^{-1})(y+2) = x \mod 26$

$x \equiv 7(y+2) \pmod{26}$

$26 = 1 \times 15 + 11$

$15 = 1 \times 11 + 4$

⋮

$1 = 7 \times 15 - 4 \times 26$

extended euclidean algorithm

Z G Y B U

189 56 182 21 154 ⇐ $x = 7(y+2)$

7 4 0 21 24 ⇐ $x = 7(y+2) \pmod{26}$

H E A V Y

* Attacking the Affine Cipher

* will come to this later *

* Exercise 3.1.1

c_1=IUBBOEKHSQHVEHJUDJXEKIQDTTEBBQHI

8 20 1 1 14 4 10 7 18 16 7 21 4 7 9 20 3 9 23 4 10 8 16 3 19 19 4 1 1 16 7 8

with $k=3 \Rightarrow$

$8-3=5 \ m26 = 5 = F$, $20-3=17 \ m26 = 17 = R$, $1-3=-2 \ m26 = 24 = Y$, Y,
$14-3=11 \ m26 = 11 = L$, $4-3=1=B$, $7-3=4=E$, $10-3=7=H$

FRYYLBHE..... × make sense.

with $k=16 \Rightarrow$

$8-16=-8 \ m26 = 18 = S$, $20-16=4=E$, $1-16=-15 \ m26 = 11 = L$, L,
$14-16=-2 \ m26 = 24 = Y$, $4-16=-12 \ m26 = 14 = O$, $10-16=-6 \ m26 = 20 = U$
$7-16=-9 \ m26 = 17 = R$, $16-16=0=A$, $7-16=-9 \ m26 = 17 = R$,
$21-16=5=F$, $4-16=-12 \ m26 = 14 = O$, $7-16 \ m26 = -9 \ m26 = 17 = R$
$9-16=-7 \ m26 = 19 = T$, $20-16=4=E$, $3-16=-13 \ m26 = 13 = N$,
T, $23-16=7=H$, $4-16=-12 \ m26 = 14$, $10-16=-6 \ m26 = 20 = U$,
$8-16=-8 \ m26 = 18 = S$, A, $3-16=-13 \ m26 = 13 = N$, $19-16=3=D$, D,
$4-16=-12 \ m26 = 14 = O$, $1-16=-15 \ m26 = 11 = L$, L, A, R, S

SELL YOUR CAR FOR TEN THOUSAND DOLLARS.

this makes sense.

Exercise 3.1.1 ⇒

$c = 11p + 17 \pmod{26}$

$y = 11x + 17 \pmod{26}$

①

H	I	R	U	S	H	A	
7	8	17	20	18	7	0	
94	105	204	237	215	94	17	$\Leftarrow 11x+17$
Q	B	W	D	H	Q	R	$\Leftarrow 11x+17 \pmod{26}$

M	E	E	T	M	E
12	4	4	19	12	4
149	61	61	226	149	61
19	9	9	18	19	9
T	J	J	S	T	J

N	A	T	I	O	N	A	L
13	0	19	8	14	13	0	11
160	17	226	105	171	160	17	138
4	17	18	1	15	4	17	8
E	R	S	B	P	E	R	I

A	T	T	H	E
0	19	19	7	4
17	226	226	94	61
17	18	18	16	9
R	S	S	Q	J

G	A	L	L	E	R	Y
6	0	11	11	4	17	24
83	17	138	138	61	204	281
5	17	8	8	9	22	21
F	R	I	I	J	W	V

HIRUSHA MEET ME AT THE NATONAL GALLERY

QBWDHQR TJJS TJ RS SQJ ERSBPERI FRIIJWV

② $c \equiv 11p + 17 \pmod{26} \Rightarrow p = (11^{-1})(c-17) \pmod{26}$

gcd(11, 26) ⇒

	26	11
$26 - 11\times2 = 4$	$1 - 0 = 1$	$0 - 1\times2 = -2$
$11 - 4\times2 = 3$	$0 - 1\times2 = -2$	$1 + 2\times2 = 5$
$4 - 3\times1 = 1$	$1 + 2 = 3$	$-2 - 5\times1 = -7$

$\therefore \; 1 = 3\times26 - 7\times11$

$-7 \bmod 26 = 19 \Rightarrow p \equiv 19(c-17) \pmod{26}$ | $326 - 26\cdot 13$

$p \equiv 19c - 323 \pmod{26}$ | 15

$p = 19c - 15 \pmod{26}$

(3) NWVASPFWRAQVRSYJRXBE

13 22 21 0 18 15 5 22 17 0 16 21 17 18 24 9 17 23 1 4

N	W	V	A	S	P	F	W	R	A	Q	V	R	S	Y	J	R	X	B	E
13	22	21	0	18	15	5	22	17	0	16	21	17	18	24	9	17	23	1	4
262	433	414	15	357	300	110	433	338	15	319	414	338	357	471	186	338	452	34	91
2	17	24	15	19	14	6	17	0	15	7	24	0	19	3	4	0	10	8	13
C	R	Y	P	T	O	G	R	A	P	H	Y	A	T	D	E	A	K	I	N

the phrase is $\Rightarrow$ CRYPTOGRAPHY AT DEAKIN

* Vigenere Cipher

just like shift cipher, but has a 'key'.

HEAVYRAIN, key is CAT (2 0 19)

	H	E	A	V	Y	R	A	I	N
	7	4	0	21	24	17	0	8	13
+	2	0	19	2	0	19	2	0	19
results	9	4	19	23	24	36	2	8	32
mod 26	9	4	19	23	24	10	2	8	6
	J	E	T	X	Y	K	C	I	G

① add key values

encryption

J E T X Y K C I G

9 4 19 23 24 10 2 8 6

− 20 19 2 0 19 2 0 19

7 4 0 21 24 −9 0 8 −13

H E A V Y R A I N

Exercise 3.2.1

Exercise 3.2.1 Given the plaintext=WEAREDISCOVEREDSAVEYOURSELVES and key=DECEPTIVEDECEPTIVEDECEPTIVEDE, find the ciphertext.

W E A R E D I S C O V E R E D S A V E Y O U R S E L V E S

22 4 0 17 4 3 8 18 2 14 21 4 17 4 3 18 0 21 4 24 14 20 17 18 4 11 21 4 18

D E C E P T I V E D E C E P T I V E D E C E P T I V E D E

3 4 2 4 15 19 8 21 4 3 4 2 4 15 19 8 21 4 3 4 2 4 15 19 8 21 4 3 4

25 8 2 21 19 22 16 39 6 17 25 6 21 19 22 26 21 25 7 28 16 24 32 37 12 32 25 7 22

$39 \downarrow 13$, $26 \downarrow 0$, $28 \downarrow 2$, $32 \downarrow 6$, $37 \downarrow 11$, $32 \downarrow 6$

ZICVTWQNGRZGVTWAVZHCQYGLMGZHW.

Exercise 3.2.2.

H E R E I S H O W I T W O R K S

7 4 17 4 8 18 7 14 22 8 19 22 14 17 10 18

21 4 2 19 14 17 21 4 2 19 14 17 21 4 2 19

28 8 19 23 22 35 28 18 24 27 33 39 35 21 12 37

2 8 19 23 22 9 2 18 24 1 7 13 9 21 12 11

C I T X W J C S Y B H N J V M L

Exercise 3.2.3 Using the permutation

$$\begin{matrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 3 & 2 \end{matrix}$$

encrypt the following plaintexts:

(a) SEND MONEY TO YOUR UNCLE

SENDM ONEYT OYOUR UNCLE

MDSNE TYOEN RUOOY ELUCN

(b) TRAVEL TO THE CITY NEXT MONDAY

TRAVE LTOTH ECITY NEXTM ONDAY

EVTAR HTLOT YTEIC MTNXE YAODN

Exercise 3.2.4 Given plaintext=00101001 and key=10101100, apply the bitwise XOR to obtain the ciphertext. The **bitwise XOR** means adding the key to the plaintext mod 2, bit by bit.

Solutions

0 0 1 0 1 0 0 1
1 0 1 0 1 1 0 0

1 0 0 0 0 1 0 1 //

* Hill Cipher

Cryptosystem with matrix as the key.

* **Encrypt** $m = CAT$ with key $K = \begin{bmatrix} 2 & 4 & 1 \\ 3 & 7 & 0 \\ 1 & 1 & 7 \end{bmatrix}$

$\Rightarrow M = [2 \;\; 0 \;\; 19]$, $C = m \times K$

$$[2 \;\; 0 \;\; 19] \begin{bmatrix} 2 & 4 & 1 \\ 3 & 7 & 0 \\ 2 & 1 & 7 \end{bmatrix}$$

$[2 \;\; 0 \;\; 19][16 \;\; 1 \;\; 5] \Rightarrow Q \; B \; F$ //

* Decrypt $\Rightarrow$ ① find $\det(K)$ & check if $\gcd(\det(K), 26) = 1$

$$\det(K) = C_{21} \times 3 + C_{22} \times 7 + C_{23} \times 0$$
$$= (-1)^{2+1} \times \det\begin{pmatrix} 4 & 1 \\ 1 & 7 \end{pmatrix} \times 3 + (-1)^{2+2} \det\begin{pmatrix} 2 & 1 \\ 2 & 7 \end{pmatrix} \times 7$$
$$= -1 \times 27 \times 3 + (14 - 2) \times 7$$
$$= 3$$

& $\gcd(3, 26) = 1$ //

② find the adjugate of matrix K.

$$\mathrm{cof}(K) = \begin{bmatrix} 23 & 5 & 15 \\ 25 & 12 & 6 \\ 19 & 3 & 2 \end{bmatrix} \Rightarrow \mathrm{Adj}(K) = \begin{bmatrix} 23 & 25 & 19 \\ 5 & 12 & 3 \\ 15 & 6 & 2 \end{bmatrix}$$

ensure to (mod 26) everything before you put it in

③ with $\det(K)$ & $\mathrm{adj}(K)$, calculate inverse of K to decrypt C.

$$K^{-1} = \frac{1}{\det(K)} \times \mathrm{Adj}(K) = 3^{-1} [\mathrm{Adj}\, K]$$

find this in (mod 26) with extended euclidean algorithm

$26 = 8 \times 3 + 2 \quad \cdots \Rightarrow \quad 3^{-1} \equiv 9$

$$\therefore K^{-1} = 9 \cdot \begin{bmatrix} 23 & 25 & 19 \\ 5 & 12 & 3 \\ 15 & 6 & 2 \end{bmatrix} = \begin{bmatrix} 23 \cdot 9 & 25 \cdot 9 & 19 \cdot 9 \\ 5 \cdot 9 & 12 \cdot 9 & 3 \cdot 9 \\ 15 \cdot 9 & 6 \cdot 9 & 2 \cdot 9 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 25 & 17 & 15 \\ 19 & 4 & 1 \\ 5 & 2 & 18 \end{bmatrix}$$

we know that $C = M \times K$

if we $\times K^{-1} \Rightarrow CK^{-1} = M \times \overbrace{K \times K^{-1}}^{I}$

$M = CK^{-1}$

$$\begin{bmatrix} 25 & 17 & 15 \\ 19 & 4 & 1 \\ 5 & 2 & 18 \end{bmatrix}$$

$[16 \ 1 \ 5][2 \ 0 \ 19]$ again, take mod 26 of actual values.

$\downarrow$

C A T //

- Exercise 3.3.1

H	I	R	U	S	H	A	A

$[7 \ 8][17 \ 20][18 \ 7][0 \ 0]$, $M = \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$

$HI \Rightarrow \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$

$7 + 8 \cdot 4 = 39 \ m26 = 13$

$14 + 8 \cdot 7 = 70 \ m26 = 18$

$[7 \ 8][13 \ 18]$

$HI \Rightarrow NS$

RU $\Rightarrow$ $\begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$

$[17 \quad 20]\,[19 \quad 18]$

$17 + 80 = 97 \; m26 = 19$

$17 \cdot 2 + 20 \cdot 7 = 174 \; m26 = 18$

RU $\Rightarrow$ TS

SH $\Rightarrow$ $\begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$

$[18 \quad 7]\;[20 \quad 7]$

$18 + 7 \times 4 = 46 \; m26 = 20$

$36 + 49 = 85 \; m26 = 7$

SH $\Rightarrow$ UH

AA $\Rightarrow$ $\begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$

$[0 \quad 0]\,[0 \quad 0]$

$0 + 0 = 0, \; 0 + 0 = 0$

AA $\Rightarrow$ AA

$\therefore$ HIRUSHAA $\Rightarrow$ NSTSUHAA

A Exercise 3.3.2 $\Rightarrow$

S I T | C R Y | P T O

C I H | O L I | N F O

3x3 matrix

SIT $\Rightarrow [18 \quad 8 \quad 19]$

CRY $\Rightarrow [2 \quad 17 \quad 24]$

PTO $\Rightarrow [15 \quad 19 \quad 14]$

$$\therefore A = \begin{bmatrix} 18 & 8 & 19 \\ 2 & 17 & 24 \\ 15 & 19 & 14 \end{bmatrix}$$

CIH $\Rightarrow [2 \quad 8 \quad 7]$

OLI $\Rightarrow [14 \quad 11 \quad 8]$

NFO $\Rightarrow [13 \quad 5 \quad 14]$

$$\therefore B = \begin{bmatrix} 2 & 8 & 7 \\ 14 & 11 & 8 \\ 13 & 5 & 14 \end{bmatrix}$$

from this, we know that $\Rightarrow$

each plaintext block $\Rightarrow$

$$[18 \quad 8 \quad 19]\, M = [2 \quad 8 \quad 7]$$
$$[2 \quad 17 \quad 24]\, M = [14 \quad 11 \quad 8]$$
$$[15 \quad 19 \quad 14]\, M = [13 \quad 5 \quad 14]$$

combining these equations $\Rightarrow$

$$A \times M = B$$

or

$$\begin{bmatrix} 18 & 8 & 19 \\ 2 & 17 & 24 \\ 15 & 19 & 14 \end{bmatrix} \times M = \begin{bmatrix} 2 & 8 & 7 \\ 14 & 11 & 8 \\ 13 & 5 & 14 \end{bmatrix}$$

$$A \times M = B$$

$$\times A^{-1} \Rightarrow \quad \underbrace{A^{-1} A}_{I} \times M = A^{-1} B$$

$$M = A^{-1} B \pmod{26}$$

to find $C(A) \rightarrow$

$$\begin{bmatrix} 18 & 8 & 19 \\ 2 & 17 & 24 \\ 15 & 19 & 14 \end{bmatrix}.$$

$$C_{11} = (-1)^{1+1} \det\begin{pmatrix} 17 & 24 \\ 19 & 14 \end{pmatrix} = 17 \cdot 14 - 24 \cdot 19 = -218 \bmod 26 = 16$$
$$C_{12} = -\det\begin{pmatrix} 2 & 24 \\ 15 & 14 \end{pmatrix} = -(28 - 24 \cdot 15) = 332 \bmod 26 = 20$$
$$C_{13} = +\det\begin{pmatrix} 2 & 17 \\ 15 & 19 \end{pmatrix} = 2 \cdot 19 - 17 \cdot 15 = -217 \bmod 26 = 17$$
$$C_{21} = -\det\begin{pmatrix} 8 & 19 \\ 19 & 14 \end{pmatrix} = -(8 \cdot 14 - 19 \cdot 19) = 249 \bmod 26 = 15$$
$$C_{22} = +\det\begin{pmatrix} 18 & 19 \\ 15 & 14 \end{pmatrix} = 18 \cdot 14 - 19 \cdot 15 = -33 \bmod 26 = 19$$
$$C_{23} = -\det\begin{pmatrix} 18 & 8 \\ 15 & 19 \end{pmatrix} = -(18 \cdot 19 - 8 \cdot 15) = -222 \bmod 26 = 12$$
$$C_{31} = +\det\begin{pmatrix} 8 & 19 \\ 17 & 24 \end{pmatrix} = 8 \cdot 24 - 19 \cdot 17 = -131 \bmod 26 = 25$$
$$C_{32} = -\det\begin{pmatrix} 18 & 19 \\ 2 & 24 \end{pmatrix} = -(18 \cdot 24 - 19 \cdot 2) = -394 \bmod 26 = 22$$
$$C_{33} = +\det\begin{pmatrix} 18 & 8 \\ 2 & 17 \end{pmatrix} = 18 \cdot 17 - 8 \cdot 2 = 290 \bmod 26 = 4$$

$$\therefore C(A) = \begin{bmatrix} 16 & 20 & 17 \\ 15 & 19 & 12 \\ 25 & 22 & 4 \end{bmatrix}, \quad Adj(A) = \begin{bmatrix} 16 & 15 & 25 \\ 20 & 19 & 22 \\ 17 & 12 & 4 \end{bmatrix}$$

$$\det(A) = 18C_{11} + 8C_{12} + 19C_{13}$$

$$= 18\cdot16 + 8\cdot20 + 19\cdot17 = 771 \bmod 26 = 17$$

$\gcd(17, 26) \Rightarrow 1$

	17	26
$26 - 17\times1 = 9$	$0 - 1 = -1$	$1 - 0 = 1$
$17 - 9\times1 = 8$	$1 - -1 = 2$	$0 - 1 = -1$
$9 - 8\times1 = 1$	$-1 - 2 = -3$	$1 - -1 = 2$

$$\therefore\ 1 = 2\times26 - 3\times17$$

$$\therefore\ -3\cdot17 \equiv 1 \pmod{26} \Rightarrow 17^{-1} \equiv -3 \pmod{26}$$

$$17^{-1} \equiv 23 \pmod{26}$$

$$\therefore\ A^{-1} = 23\times\begin{bmatrix}16 & 15 & 25\\ 20 & 19 & 22\\ 17 & 12 & 4\end{bmatrix} \pmod{26}$$

$$A^{-1} = \begin{bmatrix}368 & 345 & 575\\ 460 & 437 & 506\\ 391 & 276 & 92\end{bmatrix} \pmod{26}$$

$$A^{-1} = \begin{bmatrix}4 & 7 & 3\\ 18 & 21 & 12\\ 1 & 16 & 14\end{bmatrix} \pmod{26}$$

$$M = A^{-1}\times B$$

$\rightarrow$ next page

$$\begin{bmatrix} 2 & 8 & 7 \\ 14 & 11 & 8 \\ 13 & 5 & 14 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 7 & 3 \\ 18 & 21 & 12 \\ 1 & 16 & 14 \end{bmatrix} \begin{bmatrix} 15 & 20 & 22 \\ 18 & 19 & 20 \\ 18 & 20 & 19 \end{bmatrix}$$

$$M = \begin{bmatrix} 15 & 20 & 22 \\ 18 & 19 & 20 \\ 18 & 20 & 19 \end{bmatrix}$$

$4 \cdot 2 + 7 \cdot 14 \cdot 3 \cdot 13 = 145 \ m26 = 15$

$4 \cdot 8 + 7 \cdot 11 + 3 \cdot 5 = 124 \ m26 = 20$

$4 \cdot 7 + 7 \cdot 8 + 3 \cdot 14 = 126 \ m26 = 22$

$18 \cdot 2 + 21 \cdot 14 + 12 \cdot 13 = 486 \ m26 = 18$

$18 \cdot 8 + 21 \cdot 11 + 12 \cdot 5 = 435 \ m26 = 19$

$18 \cdot 7 + 21 \cdot 8 + 12 \cdot 14 = 462 \ m26 = 20$

$1 \cdot 2 + 16 \cdot 14 + 14 \cdot 13 = 408 \ m26 = 18$

$1 \cdot 8 + 16 \cdot 11 + 14 \times 5 = 254 \ m26 = 20$

$1 \cdot 7 + 16 \cdot 8 + 14 \cdot 14 = 331 \ m26 = 19$